

## VRATZA, Bulgaria

WMO: 155050

Lat: 43.256N

Lon: 23.533E

Elev: 310

StdP: 97.66

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -11.6      | -8.8 | -14.1                         | 1.1 | -11.0 | -11.2 | 1.5 | -7.8 | 14.3                  | 7.3  | 11.9 | 7.4  | 1.2                      | 140  | 0.432 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.3                   | 33.7            | 21.6 | 31.9 | 21.0 | 30.2 | 20.5 | 23.0                | 31.1 | 22.0 | 29.8 | 21.2 | 28.7 | 2.4                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.2                            | 15.5 | 27.7 | 19.3 | 14.6 | 26.4 | 18.5 | 13.9 | 25.1 | 69.7          | 31.1 | 65.9 | 30.0 | 62.7 | 28.8 | 29.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.6          | 7.4  | 5.9 | -14.4                      | 36.8 | 3.3                | 2.6 | -16.8                                              | 38.7 | -18.8      | 40.3 | -20.7      | 41.7 | -23.1      | 43.6 |    |    |
| (5)               |      |     | -14.8                      | 25.1 | 3.3                | 1.6 | -17.2                                              | 26.3 | -19.1      | 27.2 | -21.0      | 28.1 | -23.4      | 29.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.2   | 0.5   | 2.5   | 7.2   | 12.2  | 17.3  | 20.9  | 23.3  | 23.1  | 18.3  | 12.4  | 6.8   | 1.4  |
|                                                                                  | DBStd       | 9.22   | 5.51  | 6.00  | 5.33  | 3.92  | 3.63  | 3.45  | 3.16  | 3.22  | 3.77  | 4.52  | 4.76  | 5.31 |
|                                                                                  | HDD10.0     | 1071   | 297   | 215   | 119   | 22    | 1     | 0     | 0     | 0     | 0     | 28    | 120   | 270  |
|                                                                                  | HDD18.3     | 2717   | 553   | 442   | 344   | 188   | 63    | 14    | 2     | 3     | 46    | 191   | 346   | 523  |
|                                                                                  | CDD10.0     | 1879   | 2     | 6     | 33    | 87    | 226   | 327   | 412   | 407   | 248   | 101   | 24    | 5    |
|                                                                                  | CDD18.3     | 483    | 0     | 0     | 1     | 3     | 31    | 91    | 156   | 152   | 44    | 6     | 1     | 0    |
|                                                                                  | CDH23.3     | 4231   | 0     | 0     | 4     | 28    | 219   | 768   | 1467  | 1372  | 332   | 40    | 1     | 0    |
|                                                                                  | CDH26.7     | 1539   | 0     | 0     | 0     | 4     | 38    | 251   | 593   | 552   | 96    | 4     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.1    | 2.0   | 2.3   | 2.7   | 2.5   | 2.1   | 2.2   | 2.2   | 2.0   | 1.9   | 1.7   | 1.9   | 1.9  |
| Precipitation                                                                    | PrecAvg     | 754    | 44    | 41    | 55    | 74    | 95    | 100   | 75    | 55    | 56    | 51    | 58    | 53   |
|                                                                                  | PrecMax     | 1103   | 140   | 91    | 148   | 176   | 285   | 234   | 184   | 172   | 185   | 154   | 142   | 118  |
|                                                                                  | PrecMin     | 395    | 1     | 3     | 17    | 8     | 12    | 22    | 11    | 1     | 6     | 1     | 10    | 3    |
|                                                                                  | PrecStd     | 143    | 30    | 20    | 29    | 37    | 52    | 52    | 45    | 46    | 49    | 36    | 33    | 30   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 15.9  | 18.5  | 23.1  | 27.0  | 29.8  | 33.7  | 37.7  | 36.1  | 32.4  | 27.1  | 21.9  | 18.4 |
|                                                                                  |             | MCWB   | 9.3   | 10.6  | 13.5  | 16.2  | 19.6  | 22.2  | 21.2  | 22.5  | 20.3  | 18.0  | 14.2  | 11.9 |
|                                                                                  | 2%          | DB     | 13.0  | 15.6  | 20.2  | 22.9  | 27.5  | 31.4  | 34.0  | 33.6  | 29.3  | 24.0  | 19.2  | 15.3 |
|                                                                                  |             | MCWB   | 7.9   | 9.4   | 11.8  | 14.6  | 18.8  | 21.5  | 21.6  | 21.5  | 19.1  | 16.7  | 13.1  | 9.6  |
|                                                                                  | 5%          | DB     | 10.4  | 13.6  | 17.9  | 20.6  | 25.7  | 29.5  | 31.9  | 31.7  | 27.0  | 21.6  | 16.4  | 12.0 |
|                                                                                  |             | MCWB   | 6.5   | 8.3   | 10.8  | 13.3  | 17.9  | 20.7  | 20.9  | 21.0  | 18.0  | 15.6  | 11.3  | 7.7  |
|                                                                                  | 10%         | DB     | 8.3   | 11.5  | 15.7  | 18.8  | 24.0  | 27.7  | 30.0  | 30.0  | 24.8  | 19.2  | 13.6  | 8.7  |
|                                                                                  |             | MCWB   | 5.1   | 7.1   | 9.7   | 12.4  | 17.0  | 20.0  | 20.5  | 20.3  | 17.2  | 14.3  | 10.2  | 5.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 10.0  | 11.2  | 14.5  | 17.5  | 20.9  | 23.9  | 24.1  | 24.2  | 21.4  | 18.9  | 14.8  | 12.4 |
|                                                                                  |             | MCDB   | 15.0  | 17.2  | 21.8  | 24.8  | 28.2  | 30.9  | 32.6  | 32.9  | 29.6  | 25.9  | 20.7  | 17.8 |
|                                                                                  | 2%          | WB     | 8.2   | 9.7   | 12.3  | 15.1  | 19.5  | 22.6  | 22.7  | 23.0  | 19.6  | 17.4  | 13.3  | 9.7  |
|                                                                                  |             | MCDB   | 12.5  | 15.2  | 18.7  | 21.4  | 26.1  | 29.8  | 31.0  | 31.4  | 27.5  | 22.8  | 18.2  | 14.7 |
|                                                                                  | 5%          | WB     | 6.7   | 8.5   | 11.1  | 14.0  | 18.5  | 21.4  | 21.8  | 21.8  | 18.7  | 16.0  | 11.9  | 7.9  |
|                                                                                  |             | MCDB   | 10.1  | 13.2  | 17.7  | 19.5  | 24.8  | 28.3  | 30.1  | 29.5  | 25.8  | 20.5  | 15.4  | 11.9 |
|                                                                                  | 10%         | WB     | 5.2   | 7.1   | 9.9   | 13.0  | 17.6  | 20.3  | 21.0  | 20.8  | 17.7  | 14.7  | 10.5  | 6.2  |
|                                                                                  |             | MCDB   | 7.8   | 11.2  | 15.4  | 17.7  | 23.2  | 26.8  | 28.7  | 28.4  | 23.6  | 18.7  | 13.4  | 8.5  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.1   | 7.1   | 8.1   | 8.6   | 9.3   | 9.6   | 10.3  | 10.4  | 9.3   | 8.0   | 6.7   | 5.7  |
|                                                                                  |             | MCDBR  | 10.2  | 10.8  | 11.1  | 11.5  | 11.7  | 12.3  | 13.3  | 12.9  | 12.7  | 11.9  | 11.3  | 10.2 |
|                                                                                  |             | MCWBR  | 6.6   | 6.6   | 6.1   | 6.2   | 6.1   | 5.6   | 5.5   | 5.4   | 5.8   | 6.2   | 6.4   | 6.2  |
|                                                                                  | 5% WB       | MCDBR  | 9.8   | 10.5  | 10.9  | 10.2  | 11.0  | 11.4  | 11.9  | 11.8  | 11.6  | 11.0  | 10.3  | 9.6  |
|                                                                                  |             | MCWBR  | 6.4   | 6.7   | 6.3   | 6.0   | 6.1   | 5.9   | 5.5   | 5.7   | 5.7   | 6.2   | 6.1   | 6.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.328  | 0.356 | 0.398 | 0.456 | 0.441 | 0.438 | 0.446 | 0.458 | 0.429 | 0.386 | 0.352 | 0.329 |      |
|                                                                                  | taud        | 2.339  | 2.266 | 2.205 | 2.054 | 2.166 | 2.213 | 2.193 | 2.157 | 2.212 | 2.322 | 2.382 | 2.369 |      |
|                                                                                  | Ebn at Noon | 790    | 826   | 834   | 810   | 836   | 838   | 826   | 800   | 793   | 783   | 757   | 752   |      |
|                                                                                  | Edh at Noon | 83     | 105   | 127   | 159   | 146   | 140   | 142   | 142   | 125   | 99    | 80    | 74    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.50   | 2.32  | 3.39  | 4.57  | 5.54  | 6.22  | 6.38  | 5.71  | 4.16  | 2.75  | 1.63  | 1.25  |      |
|                                                                                  | RadStd      | 0.23   | 0.31  | 0.37  | 0.53  | 0.49  | 0.48  | 0.43  | 0.47  | 0.33  | 0.36  | 0.21  | 0.18  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | +0.70 | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (46 neighbors) | +0.50 | N/A     | N/A    | N/A     | +0.41 | N/A   | N/A         | -133    | N/A     | N/A     |

Nomenclature: See separate page